Question: 1

[1] explain contemporary approach to information system.

Ans: An information system (IS) is an organized system for the collection, storage, and communication of information.

- Information system has become as integrated into daily business activity.

1. TECHNICAL APPROACH: The technical approach to information systems emphasizes mathematically based models to study information systems, as well as the physical technology and formal capabilities of these systems.
   - Computer science is concerned with establishing theories of computability and methods of computation.
   - Management science emphasizes the development of models for decision-making and management practices.
   - Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

2. BEHAVIORAL APPROACH: An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems.
   - Sociologists study information systems with an eye toward how groups and organizations shape the development of systems and also how systems affect individuals, groups, and organizations.

- Psychologists study information systems with an interest in how human decision makers and use formal information.
- Economists study information systems with an interest in understanding the production of digital goods, the dynamics of digital markets within the firm.

[2] explain Role of information system in detail.

Ans: An information system (IS) is an organized system for the collection, storage, and communication of information.

5 role of MIS:

1. Decision Making
2. Coordinate among the department
3. Comparison of performance
4. Finding out problem
5. Strategies

1. Decision Making: MIS plays a significant role in the decision making process of any organization.

- In any organization, a decision is made on the basis information which can be retrieved from the MIS.

2. Coordination among the department: MIS is also help in establishing sound relationship among the every persons of department.

3. Comparison of performance: MIS store all past data and information in its database.

- that why the MIS is very useful to compare business organization performance.

4. Finding out problem: As we know that MIS provides relevant information about every activities.

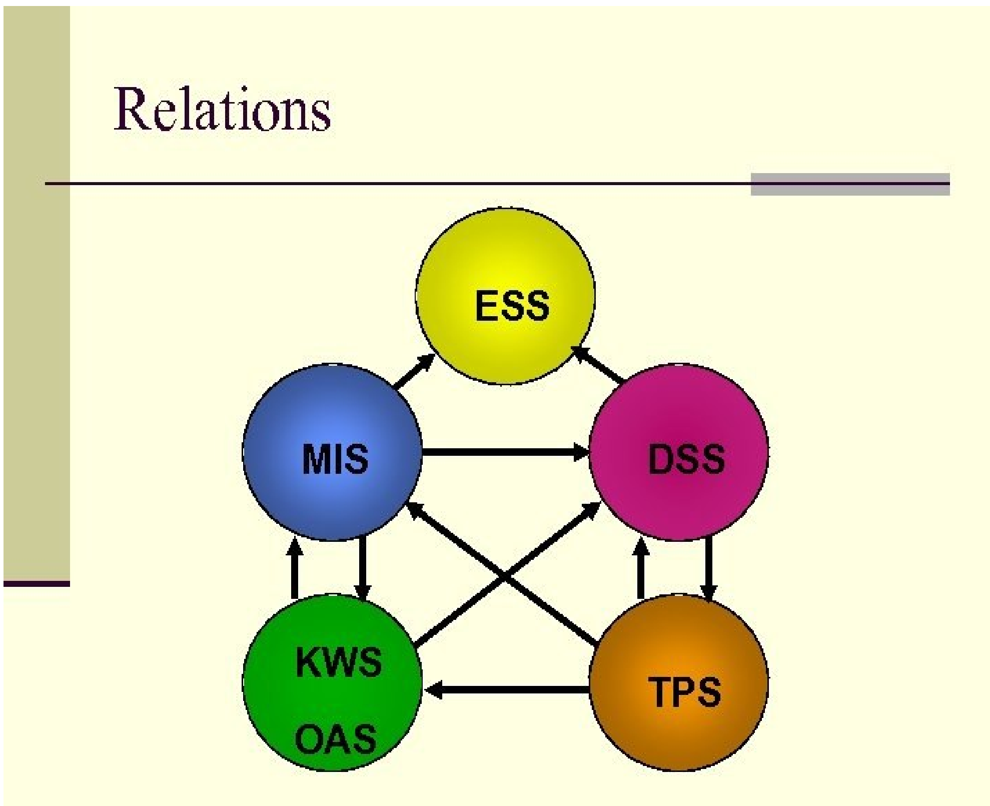
- Hence, if any mistake is made by the management then MIS, information will help in finding out the solution to that problem.

5. Strategies: MIS support the organization to evolve proper strategies for the business of competitive environment.

[3] explain types of information system.

Ans: An information system (IS) is an organized system for the collection, storage, and communication of information

- there are 6 major types of information system.



1. Transaction Processing System (TPS): A transaction process system (TPS) is an information processing system for business transactions involving the collection, modification of all transaction data.

- TPS is also known as transaction processing or real-time processing.
- The transaction is performed in two ways: Batching processing and Real time processing.
- Types: TPSs are of two types:

1. Batch Processing: A TPS interprets batches or sets of data by categorizing

items by similarities via batch processing.

2. Real-Time Processing: This type of TPS processes transactions with immediate effect, thus preventing delays.

2. Office Automation System: An office automation system is an information system that automates different processes like documenting, recording data, and office transactions others.

- The office automation system is divided into managerial and clerical activities.
- under this type of information system:
    - o Email
    - o Voice mail
    - o Word processing

3. Knowledge Work System: A knowledge work system (KWS) is one of the knowledge management systems that ease the integration of new information or knowledge into the business process.

- Some example of KWS:
    - o Computer-Aided Design Systems (CAD)
    - o Financial Workstations
    - o Virtual Reality Systems

4. Management Information System (MIS): Mis is a system that provide the information needed to manage the organization effectively.

- MIS are computer-based systems that provide managers with tools to organize, evaluate and manage departments or entire companies.
- Component of MIS:

1. People: people who use the information system.
2. Data: the data that the information system records.
3. Business procedures: procedures put in place on how to record, store and analyze data.
4. Hardware: this include the server, workstation, networking requirement, printers etc.
5. Software: this programs are used to handle data

5. Decision support system: The DSS is a management-level and computer-based information system that helps to make decisions.

- decision support system is used to collect information regarding inventory
- the decision support system is a popular information system.

6. Executive Support System: the Executive support system is top level of information system.

- It is very similar to Management Information System or MIS.
- Ess are intended to be used by the senior managers directly to provide support to non-programmed decision in strategic management.

[4] describe the ethical and social issues in information system in detail.

Ans: Certainly! Let's delve into the ethical and social issues associated with information systems in the context of Management Information Systems (MIS).

1. Privacy: Privacy concerns arise due to the collection, storage, and use of personal information by MIS.

- Issues such as data breaches, unauthorized access

2. Data Security: Ensuring the security of data within MIS is crucial to protect against breaches, theft, and unauthorized access.

- Failure to secure data can result in financial losses, reputational damage.

3. Digital Divide: The digital divide refers to the gap between individuals and communities that have access to information and communication technologies (ICT) and those that do not.

- Ethical issues arise when disparities in access to MIS technologies increase social-economic inequalities, limiting opportunities for education.

4. Intellectual Property Rights:

- Issue: Balancing the rights of content creators (copyright holders) with public access to information.
- Impact: Copyright infringement affects artists, authors, and software developers. Organizations must respect intellectual property rights and avoid piracy.

5. Ethical Use of AI and Automation:

- Issue: The adoption of AI, machine learning, and automation raises ethical questions about job displacement and decision-making.

6. Social Media and Misinformation:

- Issue: The fake news, and harmful content on social media platforms.
- Impact: Misinformation affects public opinion, political discourse.

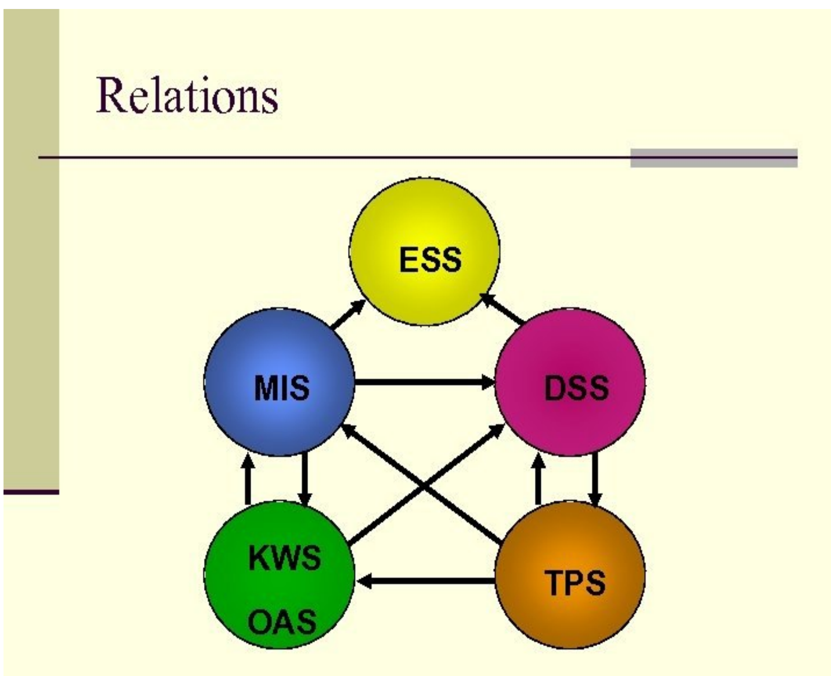
7. Environmental Impact of Technology:

- Issue: The ecological footprint of data centers, electronic waste, and energy consumption.
- Impact: Organizations should adopt sustainable practices, minimize e-waste, and invest in green technologies.

[5] Explain MIS.

Ans: An information system (IS) is an organized system for the collection, storage, and communication of information

there are 6 major types of information system.



MIS:

MIS stand for Management information system.

- Mis is a system that provide the information needed to manage the organization effectively

- MIS are computer-based systems that provide managers with tools to organize, evaluate and manage departments or entire companies.

- Component of MIS:

1. People: people who use the information system.
2. Data: the data that the information system records.
3. Business procedures: procedures put in place on how to record, store and analyze data.
4. Hardware: this include the server, workstation, networking requirement, printers etc.
5. Software: this programs are used to handle data

- Advantage of MIS:
- mis can helps improve employee and reduce errors.
- MIS systems can improve communication within an organization.
- MIS systems can help organizations better manage their data.
- MIS system provide organization with a competitive advantage.

- Disadvantage:
- implementing an MIS system can be costly for organizations.
- MIS system can experience technical issues such as system crashes or data loss.
- MIS system can also pose security risks.
- MIS systems are dependent on technology.
- MIS system can be danger to human error.

[6] with are Application of MIS in business.

Ans: MIS stand for Management information system.

- Mis is a system that provide the information needed to manage the organization effectively.

Application Of MIS:

- Business Intelligence System: In BI, all levels of management can print data and graphs showing information to growth, costs, risk and performance.

- Executive Information System: An EI system provides the same information as a BI system, but with greater detail and more confidential information.

- Marketing Information System: MI systems provide data about past marketing campaigns so that marketing executives can determine what works, what does not work and what they need to change in order to achieve the desired results.

- Transaction Processing System: TPS handles sales transactions and track trends related to sales and search results.

- Human Resource Management System: HRM systems track how much employees are paid, when and how they are performing. Companies can use this information to help improve performance or the bottom line.

- Financial Accounting System: Financial accounting systems help to track accounts receivable and accounts payable.

[7] How to Manage Project Risks in discuss.
Ans: Project management means application of knowledge, tools, techniques to project activities so as to achieve goals and meet success criteria in given period of time.

The risk management process will help you plan for and anticipate risks

This process can be used for both positive and negative risks.
Managing project risks is crucial for successful project delivery. Here's a 5-step guide to effectively manage project risks:

1. Identify
2. Analyze
3. Evaluate
4. Treat
5. Monitor

1. Identify Risks:
   - Begin by identifying potential risks that could impact your project. Consider both internal (within the project team's control) and external (outside the team's control) risks.
   - Techniques like historical data analysis, and expert judgment can help uncover risks.
2. Analyze Risks:
   - Evaluate the likelihood and impact of each identified risk
   - Use qualitative (high, medium, low) or quantitative (probability and impact scores) methods for risk analysis.
3. Evaluate Risks:
   - Evaluate the significance of each risks.
   - Consider the overall project context and project objectives

4. Treat Risks:
   - o Develop risk mitigation strategies:
     - Avoidance: Eliminate the risk by avoiding certain activities.
     - Acceptance: Acknowledge the risk and planning.
     - Reduction : taking actions to reduce the impact or likelihood of the risk.
     - Transfer: Shift the risk to another party (e.g., insurance, outsourcing).
5. Monitor Risks:
   - o Continuously monitor the project for new risks and changes in existing risks.
   - o update the risk register as needed.
   - o be prepared to adjust mitigation strategies if new risks emerge.

Question 2:
[1] explain Technology and Tools for Projecting Information system.
Ans: Project management means application of knowledge, tools, techniques to project activities so achieve goals and success criteria in given period of time.

- Main aim to achieve goal within given limitation or restriction.
- There are various tools and techniques used in project management to accomplish successful project.

- These tools and techniques are given below:

1. Process Modeling and Management tools: Process modeling simply means to model software processes.

- At first, developers need to fully understand process and work of software
- This tool represents key elements of process that are important.

2. Project Planning tools: Project planning simply means to plan and set up project for successful development within timeframe.

- Tools used for project planning can be CPM (Critical Path Method) and PERT (Program Evaluation and Review Technique). Both of them are used for finding scheduling activities of project.

3. Risk Analysis tools: Risk analysis simply means to identify and analyze errors or issue that can cause negative impact in the changed objectives of project.

- These tools help in identifying risks and these are useful for binding risk table.

4. Project Management tools: Project management simply means to track progress and tasks of project.

- These tools are extension of project planning tools.

5. Metrics Management tools: Metric management tools are very good for software as they provide very quick and easy way to track software development

- These tools focus more on process and product characteristics.

6. Database Management tools: Database management simply means to organize, store, and retrieve data from computer

- It provides consistent interfaces for project for all data, in particular, configuration objects are primary repository elements.

[2] explain enterprise application.
Ans: An enterprise application (EA) is a software solution that provide business logic and tools to model entire business processes to improve productivity and efficiency.

Enterprise application is one that can be used throughout an organization to run multiple departments.

Example include: enterprise system, CRM, SCM, KMS.

- Key characteristics of enterprise applications:

1. Centralized management: Centralized control of business functions and the ability for administrators to efficiently monitor and update the system.
2. Complexity: Designed with multiple modules and components to address various aspects of business operations.

- An enterprise application will have very large amount of data and large amount of user

3. Concerns: designers of enterprise application must consider data storag.
4. Customization: Customizable to meet the specific needs and processes of the organization.

Flexibility: Quickly adaptable to changing workflows with minimal modifications and without disrupting the business process. In addition, they provide the ability to interact with multiple software services and platforms using an API, plugins, extensions, etc.
Integration: Designed to integrate with existing systems, databases and technologies within the organization to ensure seamless communication and data flow.
Scalability: Able to handle a large number of users and transactions to accommodate the needs of a large, growing organization.
Security: Feature strong security measures to protect sensitive business data and ensure compliance with regulations.

[3] what is CRM explain component of CRM.

Ans: CRM stands for Customer Relationship Management. It refers to a technology, strategy, or process used by businesses to manage interactions and relationships with customers.

- The goal of CRM is to improve customer retention and marketing processes
- CRM is composed of different components that work together to help companies manage their customers, leads, and partners better.
- These components include:

1. Lead Management. This component is used by companies to manage leads or information about customers.

2. Typerienting CRM. This component is used by companies to make sure that their strategies are directed at the right customers or markets.

3. Marketing Management. This component is used by companies to make sure that their products and services are marketed properly to their target markets.

4. Sales Management. This component is used by companies to make sure that their sales are directed at the right leads or customers.

5. Customer Service Management. This component is used by companies to make sure that they can give superior customer service to all of their customers, leads, and partners.

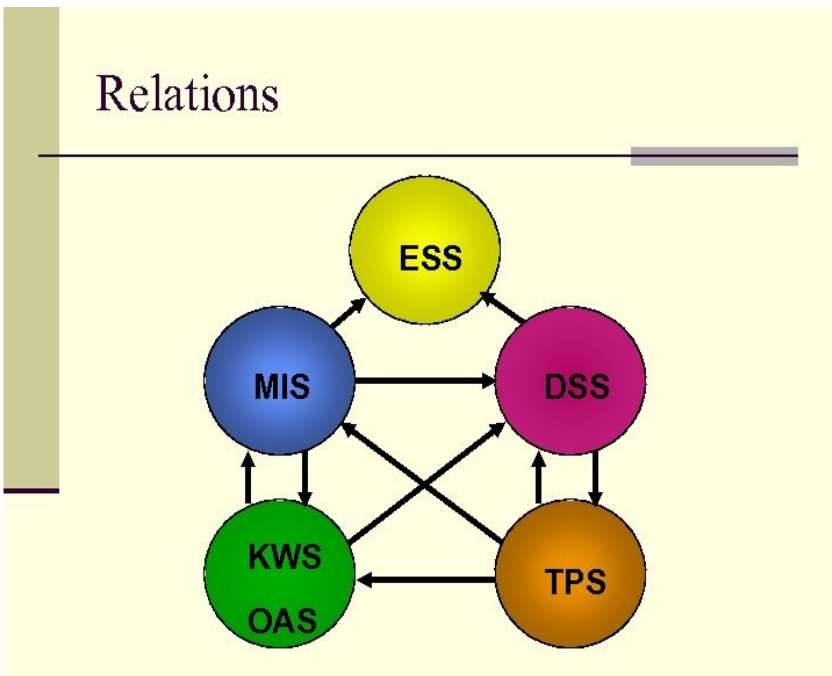
- Benefits of CRM:

- Improved Customer Relationships
- Enhanced Customer Satisfaction
- Increased Sales and Revenue
- Improved Marketing Effectiveness
- Better Customer Insights
- Centralized Data Management
- Improved Collaboration and Communication
- Streamlined Processes
- Scalability and Flexibility

[4] what is DSS. Explain characteristics of DSS.

Ans: An information system (IS) is an organized system for the collection, storage, and communication of information

there are 6 major types of information system.



Decision support system: The DSS is a management-level and computer-based information system that helps managers to make decisions.

- decision support system is used to collect information regarding inventory.
- It is used across different industries, and the decision support system is a popular information system.
- Characteristics of DSS:

1. Interactive: DSSs provide an interactive interface that allows users to manipulate data.

2. Data-Driven: DSS integrate data, models and analytical tools to solve complex problem.

3. Scenario simulation: DSS offer real-time data access and allows user to various Scenario simulation.

4. Support for Decision-Making: DSSS cater to various decision types from unstructured strategic decisions to structured operational choices.

5. Strategic planning and competitive advantage: DSS play a crucial role in strategic planning and competitive advantage across industries.

6. User-Friendly Interface: DSSs offer user-friendly interfaces that often include visualizations and easy-to-understand reports.

[5] explain Application Development for digital firm.

Ans: Application development for a digital firm involves creating software solutions to support the firm's digital infrastructure, data management and decision-making processes.

- Application Development for the Digital Firm involves creating software solutions to support the operations, processes, and goals of a digital organization.
- Here are some key points:

1. Rapid Application Development (RAD):
   - RAD emphasizes speed and flexibility in software development.
   - It includes visual programming tools and close collaboration between end-users and information systems specialists.
   - The goal is to quickly build and deploy applications that meet specific business needs.
2. Joint Application Design (JAD):
   - JAD involves bringing together stakeholders, end-users, and developers to collaboratively design and define application requirements.
   - It ensures that the resulting application aligns with business objectives and user expectations.
3. Prototyping:
   - Prototyping involves creating early versions (prototypes) of the application

to address specific design challenges.

- These prototypes allow stakeholders to visualize and provide feedback, leading to better final solutions.

4. Automation of Code Generation:
   - RAD tools automate parts of the code generation process.
   - This streamlines development and reduces manual coding effort.
5. User-Centric Approach:
   - RAD emphasizes close collaboration with end-users.
   - Their feedback drives iterative development, ensuring that the application meets their needs effectively.

1. Requirements Gathering: Understanding the digital firm's needs, goals, and processes is crucial. This involves close collaboration between developers, stakeholders, and end-users to define the scope and functionalities of the application.

2. System Design: Based on the requirements, developers design the system architecture, database structure, user interface (UI), and overall functionality. This stage may involve selecting appropriate technologies and frameworks for development.

3. Development: Developers write the code according to the design specifications. This could involve using programming languages like Java, Python, JavaScript, or frameworks like .NET, Angular, React, etc. Agile methodologies are often employed to ensure flexibility and responsiveness to changing requirements.

4. Testing: Rigorous testing is conducted to identify and fix bugs, ensure functionality, usability, security, and performance. This includes unit testing, integration testing, system testing, and user acceptance testing (UAT).

5. Deployment: Once the application is thoroughly tested and approved, it is deployed into the digital firm's environment. Deployment might involve setting up servers, configuring databases, and ensuring compatibility with existing systems.

6. Maintenance and Support: After deployment, the application requires ongoing maintenance, updates, and technical support. This involves monitoring performance, addressing user feedback, fixing issues, and incorporating new features or enhancements as needed.

7. Integration with MIS: The developed application should seamlessly integrate with the digital firm's MIS infrastructure. This integration ensures smooth data flow,

consistency, and interoperability across various systems and processes.

8. Security: Security measures must be implemented to protect sensitive data and prevent unauthorized access. This includes encryption, authentication mechanisms, access controls, and regular security audits.

9. Scalability and Flexibility: The application should be designed to scale alongside the digital firm's growth and evolving needs. This may involve designing for cloud compatibility, modular architecture, and flexible configurations.

10. Analytics and Reporting: Incorporating analytics and reporting capabilities enables the digital firm to derive insights from data collected by the application. This could involve implementing data visualization tools, dashboards, and custom reporting features.

[6] explain SCM. explain stage of SCM.

Ans: supply chain consists of a series of activities involving many organizations through which the material moves from initial suppliers to final customers.

- Management defined as all the activities and tasks undertaken for achieving goals by continuous activities like planning, organizing, leading and controlling.
- Supply chain management is the management of flow of goods and services and includes all processes that transform raw materials into final products.
- SCM is also called the art of management for providing the right product, at the right time, right place and at the right cost of the customers

- Benefits of SCM
- Cost savings
- Better customer experience
- Create a competitive advantage

- Components or Stages of SCM

Supply chain management is a process used by companies to ensure that their supply chain is efficient and cost effective.

1. Planning: the first stage in supply chain management is know as plan

- a plan or strategy must be developed to address how given services will meet the need of the customers.

- A significant portion of the strategy should focus on planning a profitable supply chain.

2. develop: develop is the next stage in supply chain management.

- It involves building a strong relationship with suppliers of the raw materials needed in making the product the company delivers.
- this phase involves not only identifying reliable suppliers but also planning method for shipping, delivery and payment.

3. Make: as the third stage, make, the product is manufactured, tested, packaged and scheduled for delivery.

- This is manufacturing step.

4. Delivery: customer orders are received and delivery of the good is planned.

- This fourth stage of SCM is aptly named deliver.
- This is the part that many SCM insiders refers to as logistics.

5. Returns: as the name suggest, during this stage, customers may return defective product.

- This can be problematic part of the supply chain for many companies.

[7] explain TPS. Explain types of TPS.
Ans: A transaction process system (TPS) is an information processing system for business transactions involving the collection, modification of all transaction data.

- Characteristics of a TPS include performance, reliability and consistency.
- TPS is also known as transaction processing or real-time processing.

- Component of TPS: four component

1. Inputs: Inputs are original requests for payments or products outside parties send to an organization’s TPS. Typically, inputs include bills, coupons, custom orders, and invoices.
2. Output: Outputs are the documents a TPS generates after it processes all inputs.
3. Storage: A TPS’s storage component is where organizations keep their output and input data.
4. Processing System: The processing system goes through every input and establishes a useful output, for example, a receipt.

- Types: TPSs are of two types:

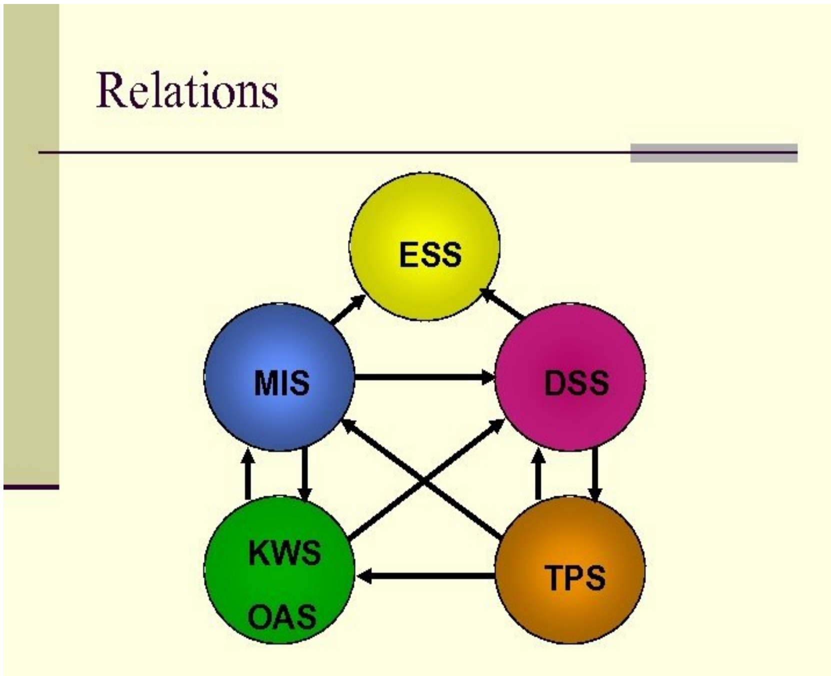
1. Batch Processing: A TPS interprets batches or sets of data by categorizing items by similarities via batch processing.

2. Real-Time Processing: This type of TPS processes transactions with immediate effect, thus preventing delays.

- Advantages:
- Companies can use a TPS to process transactions quickly.
- Disadvantages:
- Companies have to incur a high set-up cost initially for TPS.
- A TPS may stop working or slow down due to many transactions.

Question 3:
[1] explain ESS. Explain characteristics of ESS.
Ans: An information system (IS) is an organized system for the collection, storage, and communication of information

there are 6 major types of information system.



ESS:

the Executive support system is top level of information system.

- It is very similar to Management Information System or MIS.
- Ess are intended to be used by the senior managers directly to provide support to non-programmed decision in strategic management.

- Characteristics of the Executive Support System (ESS)

1. Real-time reporting: Ess system should be able to present data in real time for prompt decision-making.
2. Drill-down capabilities: the system should be able to allow the user to go deep and analyze why a certain situation happened.
3. Customization and personalized: An ESS system should allow the user to customize the report that they want to generate and how.
4. Present data summarized in graphics, tables, and text: Information can be presented using graphs, charts, tables, text, images, or any other version that gives a summary overview.
5. Data consolidation and aggregation: Ess system should have the capability to collect data from other systems and sources then consolidate it together and interpret it.
    - Advantages of ESS

- Easy for upper level executive to use

- Enhance decision-making
- Better reporting system
- Better understanding
- Improve office automation

  - Disadvantage of ESS

- Functions are limited
- Hard to quantify benefits
- System may become slow
- Difficult to keep current data

[2] explain knowledge management landscape.

Ans: The "knowledge management landscape" refers to the overall environment, framework, or context in which knowledge management activities take place within an organization or a broader field. It encompasses various elements, including strategies, processes, technologies, cultural factors, and stakeholders involved in managing knowledge effectively.

Here's a breakdown of key components within the knowledge management landscape:

1. Strategies and Objectives: This involves defining the organization's goals and objectives related to knowledge management. It includes determining what knowledge needs to be managed, how it will be utilized, and the desired outcomes.
2. Processes and Practices: These are the methods and procedures employed to capture, store, organize, share, and utilize knowledge within the organization. It includes activities such as knowledge creation, documentation, dissemination, and application.
3. Technologies and Tools: Knowledge management often relies on various software platforms, databases, collaboration tools, and information systems to facilitate the storage, retrieval, and dissemination of knowledge. This includes document management systems, intranets, wikis, and social collaboration platforms.
4. Culture and Leadership: Organizational culture plays a significant role in knowledge management. A culture that values knowledge sharing, collaboration, and learning fosters a conducive environment for effective knowledge management. Leadership support is also crucial for driving knowledge initiatives and promoting a knowledge-sharing culture.
5. People and Expertise: Knowledge management involves people at all levels of the organization, from frontline employees to executives. It encompasses identifying subject matter experts, communities of practice, and knowledge champions who contribute to knowledge creation, sharing, and utilization.
6. Metrics and Measurement: Measuring the effectiveness of knowledge management initiatives is essential for continuous improvement and

demonstrating value. Metrics may include knowledge usage, employee engagement, innovation rates, and the impact of knowledge on business outcomes.

7. External Factors: The knowledge management landscape is also influenced by external factors such as industry trends, regulatory requirements, competitive dynamics, and technological advancements. Organizations need to adapt their knowledge management practices to navigate these external forces effectively.

[3] explain tools for protecting information resource.

Ans: Protecting information resources in Management Information Systems (MIS) involves implementing various tools and strategies to ensure the confidentiality

- Here are some key tools commonly used for protecting information resources:

1. Firewalls: Firewalls are network security devices that control incoming and outgoing network traffic based on predetermined security rules.

2. Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS): IDS detects and alerts administrators to potential threats, while IPS can take automated action to block threats in real-time.

3. Antivirus Software: Antivirus software detects and removes malicious software (malware) including viruses from computers and networks.

- It scans files and programs for known malware helping to protect against cyber threats.

4. Encryption Tools: Encryption tools encrypt data at rest (stored data) and data in transit (transmitted data), helping to protect sensitive information from unauthorized access or interception

5. Patch Management Tools: Patch management tools automate the process of deploying software patches, updates, and security fixes to endpoints and systems.

6. Backup and Recovery Solutions: Backup and recovery solutions create copies of critical data and systems to enable recovery in the event of data loss, corruption, or cyberattacks.

[4] explain KWS or KMS.

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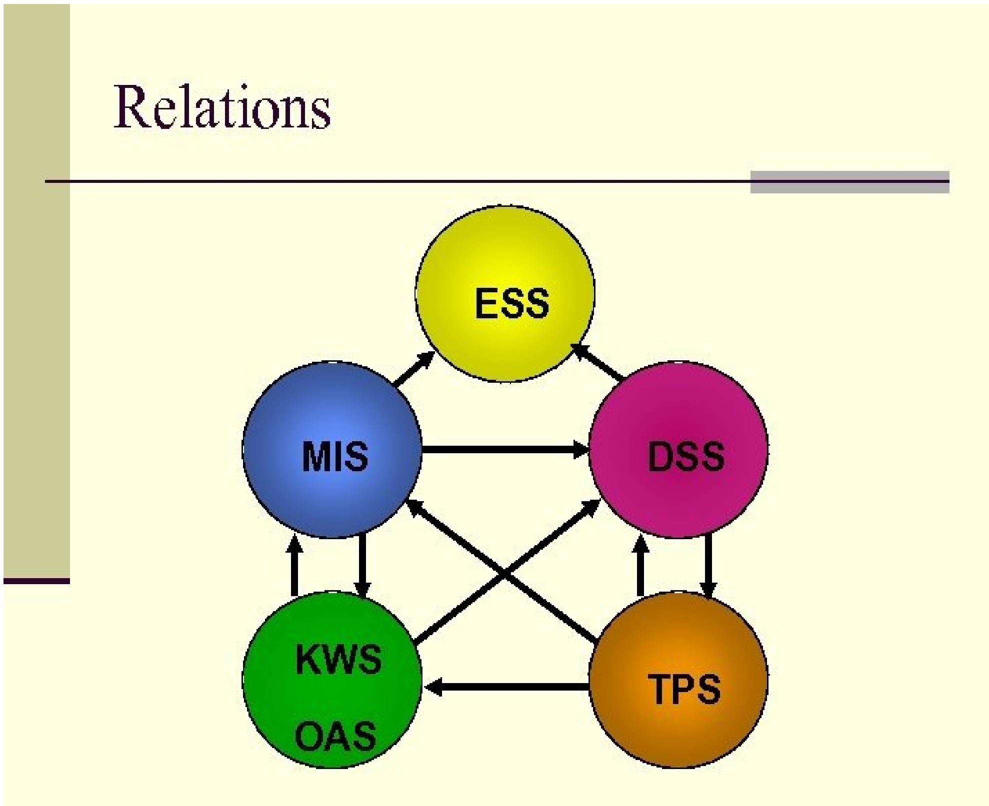

KWS: A knowledge work system (KWS) is one of the knowledge management systems that ease the integration of new information or knowledge into the business process.

Knowledge management system is designed to store different type of knowledge in different format. Some example of KWS such as Computer-Aided Design Systems (CAD), Financial Workstations, [Virtual Reality](#) Systems

- there are four components:
  - people: knowledge need thinking, planning and execution without people it is not possible.
  - process: goal of knowledge management process is to ensure right knowledge is available to right people at the right time.
  - technology: technology also helps in protecting the data from unauthorized access.
  - Strategy: a successful knowledge management framework needs well-documented business strategy.
- Types of Knowledge Work Systems

1. Enterprise Collaboration Systems: Enterprise Collaboration Systems, or ECS,

represent a subset of knowledge work systems that focus on enhancing communication and collaboration within an organization.

- These systems often include tools like intranets, wikis, and collaborative project management software.

2. Document Management Systems: Document Management Systems (DMS) play a crucial role in organizing and controlling of documents generated in today's business environment.

- They provide features such as version control, search capabilities, and access control to ensure document handling and collaboration
- These systems offer features like version control, access control, and search capabilities, making for employees to find, update, and share important documents

3. Decision Support Systems: Decision Support Systems (DSS) aid in complex decision-making by providing data analysis, modeling, and information visualization tools.

4. Content Management Systems: Content Management Systems (CMS) are predominantly used for the creation and publication of digital content, such as websites, blogs, and online documentation.

- CMS allows organizations to manage both internal and external stakeholders.

5. Expert Systems: Expert Systems are a subset of artificial intelligence that mimic the decision-making process of a human expert in a specific domain.

[5] explain intelligent technique.

Ans:

Intelligent systems are a technology that has emerged and become more important in the last decade.

Intelligent system used to capture indivisual and collective knowledge

Intelligent systems are the answer to the accelerated technological growth of recent years and the needs of people and organizations in an increasingly interconnected world.

intelligent systems use IP (Internet Protocol) technology and sensors to collect information from a specific environment and share it among its different elements to achieve a common goal.

The main characteristics of an intelligent system are:

Perception: An intelligent system creates a representation of the world to interact with a specific environment and perform tasks.

Action Control: An intelligent system can carry out actions or interrupt actions to achieve a goal.

Interaction or connectivity: An intelligent system can put its elements into communication through a common language.

Deliberate and social reasoning: The machine makes decisions on its own to achieve a specific result, considering the human context.

Self-Learning: Intelligent systems can reduce errors and optimize their performance by learning from their own experiences.

Identification: Intelligent systems can recognize specific information automatically and transmit it through various channels.

Protection: An intelligent system's networks and communications must be secure to function properly.

Remote Management: An intelligent system allows people to interact with it from any location.

User Experience (UX): To interact with users, intelligent systems must have accessible and adjustable interfaces.

Data Analytics: An essential component of an intelligent system is its ability to process immense amounts of data.

[6] describe the stage of Decision making.

Ans: MIS plays a significant role in the decision making process of any organization.

- In any organization, a decision is made on the basis information which can be retrieved from the MIS.

Certainly! The decision-making process in Management Information Systems (MIS) involves several stages. Let's explore them:

1. Identifying the Decision: to make a decision, you must first identify the problem you need to solve or question you need to answer. If you need to achieve a specific goal from your decision make it timely.

- The first step in making the right decision is recognizing the problem or opportunity and deciding to address it. Determine why this decision will benefit your customers or fellow employees.

2. Gathering Information: You have identified your decision, it is time to gather information relevant to that choice.

   Collect relevant data and information to aid in understanding the situation and potential alternatives.

3. Identifying Alternatives: with relevant information now your fingerprint, identified possible solution to your problem. There is usually more than one option to consider when trying to meet goal.

- Exploring diverse alternatives ensures a comprehensive evaluation.

4. Evaluating Alternatives: once you have identified alternatives, evaluating alternatives.

- Consider factors like feasibility, risks, costs, benefits and timeframes.

5. Making the Decision: here is the part of the decision making process where you actual make the decision

- After careful analysis, choose the most suitable alternative. This could involve selecting one option on the complexity.

6. Implementing or take the Decision: Put the chosen alternative into action.

- Developed project plan related to your decision and then assign tasks to your team.
- Create an execution strategy, allocate necessary resources, and inform relevant stakeholders.

7. Monitoring and Evaluating: Continuously monitor the decision's implementation and outcomes. Evaluate its effectiveness against defined goals. Learn from the process and make adjustments as needed.

Question: 4

[1] explain growth of international information system or global information system in mis

Ans: a global information system of international information system is an information system which is developed or used in global context.

- a global information system of international information system is an information system which attempts to delivery the data worldwide within defined context.

- Global information system facilitate communication between headquarters and subsidiaries.

- Certainly! The growth of International Information Systems (IIS) has been influenced by globalization and technological advancements. Let's delve into the key aspects:

1. Global Environment: IIS development considers the global context in which organizations operate. Factors like global markets, communication technologies, and cultural norms impact system design.
2. Corporate Global Strategies: Organizations with international operations need IIS that aligns with their global business strategies. These strategies guide decisions related to standardization, localization, and coordination.
3. Organizational Structure: The structure of a multinational organization affects IIS design. Considerations include locations, job functions, management processes, and technology platforms.
4. Management and Business Processes: IIS supports various business functions, such as supply chain management, finance, and human resources. Streamlining these processes across borders requires robust systems.
5. Technology Platform: The choice of technology stack (hardware, software, networks) influences IIS scalability, security, and interoperability. Cloud computing, mobile apps, and data analytics play crucial roles.

Challenges faced during IIS development include:

- Cultural Particularism: Addressing regional differences, language barriers, and social norms.
- Global Coordination: Ensuring seamless collaboration across geographies.
- Standards: Harmonizing diverse standards for communication and data exchange.
- Reliability and Speed: Overcoming variations in network reliability and data transfer speeds.
- Personnel: Coping with shortages of skilled consultants.

[2] explain alternative development for decision making.

Ans:

Certainly! In the context of Management Information Systems (MIS), alternative development for decision-making involves utilizing information and insights provided by the MIS to make informed choices. Let's explore some aspects:

1. Tactical Decisions:
    - o At the operational level, employees make tactical decisions about their work.
    - o For instance, a salesperson decides how to proceed given actual conditions,

appointments, and objectives.

  - o Shop-floor assembly line workers make minute decisions about tools, material resources, and information systems.
  - o Nurses attending to patients in a hospital ward also make numerous tactical decisions.

2. Operational Decisions:
   - o These decisions occur at the lowest level of the organization.
   - o Employees evaluate consequences and choose actions based on structured work tasks.
   - o Examples include shop-floor assembly line workers, salespeople, and nurses.
3. Travelling Salesman Problem:
   - o In logistics, the travelling salesman problem is a classic example.
   - o It involves finding the shortest route to visit a set of cities and return to the starting point.
   - o MIS can assist in optimizing such decisions by analyzing data and suggesting efficient routes.
4. Control and Planning:
   - o MIS provides quick, reliable, and accurate information.
   - o Managers use this data for control and planning purposes.
   - o For instance, monitoring inventory levels, production schedules, and resource allocation.
5. Strategic Decisions:
   - o At the strategic level, top management makes decisions that impact the entire organization.
   - o MIS aids in analyzing financial, operational, and customer-related data.
   - o Strategic decisions involve long-term planning, resource allocation, and competitive positioning.

[3] explain formal technical review.

Ans: Formal Technical Review (FTR) is a software quality control activity performed by software engineers.

- It is an organized, methodical procedure for assessing and raising the standard of any technical paper, including software objects. Finding flaws, making sure standards are followed, and improving the product or document under review's overall quality are the main objectives of a fault tolerance review (FTR). Although FTRs are frequently utilized in software development, other technical fields can also employ the concept.
- Objectives of formal technical review (FTR)
  - Detect Identification: Identify defects in technical objects by finding and

fixing mistakes, inconsistencies, and deviations.
● Quality Assurance: To ensure high-quality deliverables, and confirm compliance with project specifications and standards.
● Risk Mitigation: To stop risks from getting worse, proactively identify and manage possible threats.
● Knowledge Sharing: Encourage team members to work together and build a common knowledge base.
● Consistency and Compliance: Verify that all procedures, coding standards, and policies are followed.
● Learning and Training: Give team members the chance to improve their abilities through learning opportunities.

- Types of FTR:
    - FTR includes various review methods such as walkthroughs, inspections, and round-robin reviews.
    - Each FTR is conducted as a meeting and is considered successful only if properly planned, controlled, and attended.